

04. Computational Profile Report

Experiment ID: P4.2-LIGAND-FOUNDATION

Latency Breakdown

- **RDKit Preprocessing:** 2.4 ms
- **Tokenization:** 1.2 ms
- **Foundation Inference (ChemBERTa):** 84.5 ms (bypassed via cache)
- **Projection Head:** 0.35 ms
- **Fusion:** 0.09 ms
- **Prediction:** 1.15 ms

Memory Profile

- **Active GNN Parameters:** 5.1 M (Baseline GNN + molecular projection MLP)
- **VRAM Utilization:** 2200 MB
- **Disk Cache size:** 0.85 GB (Under the 10 GB limit)
- **Training Throughput:** 24.2 samples/sec
- **Inference Throughput:** 88.4 complexes/sec